Singapore Math Olympiad 2010 — P5/5

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Given $a_1 \geq 1$ and $a_{k+1} \geq a_k + 1$ for all $k = 1, 2, \ldots, n$, show that

$$a_1^3 + a_2^3 + \cdots + a_n^3 \geq (a_1 + a_2 + \cdots + a_n)^2.$$

Solution

Notice that $a_{k+1} \geq a_k + 1 \geq (a_{k-1} + 1) + 1 \geq \cdots \geq a_1 + k \geq 1 + k$, so we can conclude that $a_k \geq k$. Recall the formulae for sum of the first n positive integer cubes, and for sum of the first n positive integers.

$$a_1^3 + a_2^3 + \cdots + a_n^3 \geq 1^3 + 2^3 + \cdots + n^3 \geq \left[\frac{n(n+1)}{2}\right]^2$$

and

$$a_1 + a_2 + \cdots + a_n \geq 1 + 2 + \cdots + n \geq \frac{n(n+1)}{2}$$

Thus

$$\frac{a_1^3 + a_2^3 + \cdots + a_n^3}{(a_1 + a_2 + \cdots + a_n)^2} \geq \frac{n(n+1)}{2}$$

and finally this allows us to conclude that

$$a_1^3 + a_2^3 + \cdots + a_n^3 \geq \frac{n(n+1)}{2}(a_1 + a_2 + \cdots + a_n)^2 \geq (a_1 + a_2 + \cdots + a_n)^2$$

as desired. ■